Fabian Gittins

Contact Information

Department of Physics and Astronomy
Johns Hopkins University, 3400 N. Charles Street
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Citizenship

United Kingdom

Research Interests

Relativistic astrophysics – Gravitational-wave astronomy – Extreme physics of neutron stars

One major theme is building predictive, physically faithful neutron-star models with realistic microphysics and dynamics. Second major theme is using gravitational waves to extract this physics and constrain dense nuclear matter. Currently, advancing gravitational-wave asteroseismology to probe dense matter through modelling and detection of neutron-star oscillation modes.

Education

PhD, Mathematics, University of Southampton, GB — **Sep 2021**

Advisor: Prof Nils Andersson

Thesis title: *Gravitational waves from deformed neutron stars: mountains and tides*

MSci, Physics, University of Birmingham, GB — **Jul 2017**

Grade: First class honours

Undergraduate Master's degree with focus on theoretical physics

Research Positions

Marie Skłodowska-Curie Fellow, Johns Hopkins University, US — **Oct 2026–Present**

Marie Skłodowska-Curie Fellow, Utrecht University, NL — **Oct 2024–Sep 2026**

Research Fellow, University of Southampton, GB — **Oct 2021–Sep 2024**

PhD Researcher, University of Southampton, GB — **Sep 2017–Sep 2021**

Honours and Awards

Marie Skłodowska-Curie Postdoctoral Fellowship (Global), EU — **Oct 2026–Sep 2029**
Project lead of *QuarkWave*; €454,103

Marie Skłodowska-Curie Postdoctoral Fellowship (European), EU — **Oct 2024–Sep 2026**
Project lead of *DynTideEOS*; €203,464

Gravitational Physics Thesis Prize, Institute of Physics, GB — **2021**

Best Publication in Gravitational Physics, University of Southampton, GB — **2021**

Physics Scholarship, University of Birmingham, GB — **2013**

Publication Summary

Full list of publications can be found on Google Scholar, INSPIRE-HEP and NASA ADS.

h-index—16 (according to Google Scholar) or 14 (according to INSPIRE-HEP). Both include collaboration papers.

Top five cited—Excluding long-author-list papers. Citation counts according to Google Scholar.

1. **Gittins, F.**, Andersson, N., Jones, D. I., *Modelling neutron star mountains*, Mon. Not. R. Astron. Soc. **500**, 5570 (2021) [arXiv:2009.12794]. (83 citations)
2. **Gittins, F.**, Andersson, N., *Modelling neutron star mountains in relativity*, Mon. Not. R. Astron. Soc. **507**, 116 (2021) [arXiv:2105.06493]. (75 citations)
3. **Gittins, F.**, Andersson, N., Pereira, J. P., *Tidal deformations of neutron stars with elastic crusts*, Phys. Rev. D **101**, 103025 (2020) [arXiv:2003.05449]. (52 citations)
4. Pereira, J. P., Bejger, M., Andersson, N., **Gittins, F.**, *Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts*, Astrophys. J. **895**, 28 (2020) [arXiv:2003.10781]. (44 citations)
5. **Gittins, F.**, *Gravitational waves from neutron-star mountains*, Class. Quantum Gravity **41**, 043001 (2024) [arXiv:2401.01670]. (37 citations)

Collaboration Publications

Since 2026, I have been a co-author of refereed LIGO-Virgo-KAGRA and Einstein Telescope Collaboration publications. I list only non-collaboration articles below.

Submitted Publications

[29] Wouters, T., Ng, T. C. K., Koehn, H., **Gittins, F.**, (3 more authors), *Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope* [arXiv:2610.07975] (submitted to Phys. Rev. Lett.).

[28] Yu, H., (6 authors), **Gittins, F.**, Nanda, A., *Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions* [arXiv:2607.07943] (submitted to Phys. Rev. D).

Accepted Publications

[27] **Gittins, F.**, Narola, H., Wouters, T., Pang, P. T. H., Hinderer, T., Van Den Broeck, C., *Detecting tidal resonances in binary neutron stars*, Phys. Rev. Lett. [arXiv:2606.06376].

Refereed Publications

Research articles

[26] Negri, L., (8 authors), **Gittins, F.**, (6 more authors), *Impact of the Einstein Telescope's duty cycle on the estimation of binary black holes parameters*, Phys. Rev. D **114**, 082003 (2026) [arXiv:2606.19201].

[25] Andersson, N., Counsell, A. R., **Gittins, F.**, Ghosh, S., *Tidal response of a relativistic star*, Phys. Rev. D **113**, 064051 (2026) [arXiv:2511.05139].

[24] Yin, S., Andersson, N., **Gittins, F.**, *A post-Newtonian approach to neutron star oscillations*, Class. Quantum Gravity **42**, 235002 (2025) [arXiv:2504.06918].

[23] Pnigouras, P., Andersson, N., **Gittins, F.**, Counsell, A. R., *Dynamical neutron star tides: the signature of a mode resonance*, Mon. Not. R. Astron. Soc. **542**, 1375 (2025) [arXiv:2508.06416].

[22] Counsell, A. R., **Gittins, F.**, Andersson, N., Tews, I., *Interface modes in inspiralling neutron stars: A gravitational-wave probe of first-order phase transitions*, Phys. Rev. Lett. **135**, 081402 (2025) [arXiv:2504.06181].

[21] **Gittins, F.**, Andersson, N., Yin, S., *Perturbation theory for post-Newtonian neutron stars*, Class. Quantum Gravity **42**, 135014 (2025) [arXiv:2503.03345].

[20] **Gittins, F.**, Andersson, N., *Neutron-star seismology with realistic, finite-temperature nuclear matter*, Phys. Rev. D **111**, 083024 (2025) [arXiv:2406.05177].

[19] **Gittins, F.**, Matur, R., Andersson, N., Hawke, I., *Problematic systematics in neutron-star merger simulations*, Phys. Rev. D **111**, 023049 (2025) [arXiv:2409.13468].

[18] Counsell, A. R., **Gittins, F.**, Andersson, N., Pnigouras, P., *Neutron star g modes in the relativistic Cowling approximation*, Mon. Not. R. Astron. Soc. **536**, 1967 (2025) [arXiv:2409.20178].

[17] Counsell, A. R., **Gittins, F.**, Andersson, N., *The impact of nuclear reactions on the neutron-star g-mode spectrum*, Mon. Not. R. Astron. Soc. **531**, 1721 (2024) [arXiv:2310.13586].

[16] Pnigouras, P., **Gittins, F.**, Nanda, A., Andersson, N., Jones, D. I., *The dynamical tides of spinning Newtonian stars*, Mon. Not. R. Astron. Soc. **527**, 8409 (2024) [arXiv:2205.07577].

[15] Beri, A., (5 authors), **Gittins, F.**, Celora, T., *AstroSat and NuSTAR observations of XTE J1739-285 during the 2019-2020 outburst*, Mon. Not. R. Astron. Soc. **521**, 5904 (2023) [arXiv:2303.13085].

[14] **Gittins, F.**, Celora, T., Beri, A., Andersson, N., *Modelling neutron-star ocean dynamics*, Universe **9**, 226 (2023) [arXiv:2304.05413].

[13] **Gittins, F.**, Andersson, N., *The r-modes of slowly rotating, stratified neutron stars*, Mon. Not. R. Astron. Soc. **521**, 3043 (2023) [arXiv:2212.04892].

[12] Andersson, N., **Gittins, F.**, *Formulating the r-mode problem for slowly rotating neutron stars*, Astrophys. J. **945**, 139 (2023) [arXiv:2212.04837].

[11] Andersson, N., **Gittins, F.**, Yin, S., Panosso Macedo, R., *Building post-Newtonian neutron stars*, Class. Quantum Gravity **40**, 025016 (2023) [arXiv:2209.05871].

[10] Riley, J., (6 authors), **Gittins, F.**, (16 more authors), *Rapid stellar and binary population synthesis with COMPAS*, Astrophys. J. Suppl. Ser. **258**, 34 (2022) [arXiv:2109.10352].

[9] **Gittins, F.**, Andersson, N., *Modelling neutron star mountains in relativity*, Mon. Not. R. Astron. Soc. **507**, 116 (2021) [arXiv:2105.06493].

[8] **Gittins, F.**, Andersson, N., Jones, D. I., *Modelling neutron star mountains*, Mon. Not. R. Astron. Soc. **500**, 5570 (2021) [arXiv:2009.12794].

[7] **Gittins, F.**, Andersson, N., Pereira, J. P., *Tidal deformations of neutron stars with elastic crusts*, Phys. Rev. D **101**, 103025 (2020) [arXiv:2003.05449].

[6] Pereira, J. P., Bejger, M., Andersson, N., **Gittins, F.**, *Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts*, Astrophys. J. **895**, 28 (2020) [arXiv:2003.10781].

[5] **Gittins, F.**, Andersson, N., *Population synthesis of accreting neutron stars emitting gravitational waves*, Mon. Not. R. Astron. Soc. **488**, 99 (2019) [arXiv:1811.00550].

Review articles

[4] **Gittins, F.**, *Gravitational waves from neutron-star mountains*, Class. Quantum Gravity **41**, 043001 (2024) [arXiv:2401.01670].

Software articles

[3] Riley, J., (6 authors), **Gittins, F.**, (16 more authors), *COMPAS: A rapid binary population synthesis suite*, J. Open Source Softw. **7**, 3838 (2022).

Conference proceedings

[2] Thomas, A., Stevenson, E., **Gittins, F.**, (5 more authors), *Galactic archaeology with TESS: Prospects for testing the star formation history in the solar neighbourhood*, EPJ Web Conf. **160**, 05006 (2017) [arXiv:1610.08862].

White Papers

[1] Dietrich, T., (19 authors), **Gittins, F.**, (22 more authors), *ESO Expanding Horizon White Paper: Revealing the properties of matter at supranuclear densities with gravitational waves* [arXiv:2512.16971].

Invited Talks

14. *Physics Colloquium* — **7 Oct 2026**
University of Maryland, Baltimore County, US
13. *Astrophysics Talks Series* — **2 Apr 2026**
Istanbul University Observatory, TR (online)
12. *International Research Network for Nuclear Astrophysics Seminar* — **16 Jan 2026**
(online)
11. *CoCoNuT Meeting 2025* — **28 Oct 2025**
University of Strasbourg, FR
10. *High Energy Particle Physics and Cosmology Theory Seminar* — **30 Sep 2025**
Johns Hopkins University, US
9. *Institute for Nuclear Theory Program 25-2b* — **17 Sep 2025**
University of Washington, US
8. *Gravitational Wave Meeting* — **18 Jun 2025**
National Institute for Subatomic Physics, NL (online)
7. *Astrophysics Seminar* — **30 May 2024**
Mullard Space Science Laboratory, University College London, GB
6. *Gravitational Wave Group* — **14 Dec 2023**
Institute of Cosmology and Gravitation, University of Portsmouth, GB
5. *SPINS-UK Seminar* — **7 Jun 2023**
(online)
4. *Symposium on Gravitational Wave Physics and Astronomy: Genesis* — **28 Apr 2022**
Kyoto University, JP (online)
3. *22nd BritGrav Conference* — **5 Apr 2022**
University of Glasgow, GB (online)
2. *Colloquium* — **6 Oct 2020**
Albert Einstein Institute, Hannover, DE (online)

1. *LIGO-Virgo Collaboration Continuous Waves Working Group* **5 Dec 2018**
   (online)

Contributed Talks (Selected)

27 contributed talks at 25 separate conferences and meetings, including:

10. *NNV section for (astro)particle physics fall meeting* **7 Nov 2025**
    Soesterberg, NL
9. Joint *24th International Conference on General Relativity and Gravitation* **17 Jul 2025**
   and *16th Edoardo Amaldi Conference on Gravitational Waves*, Glasgow, GB
8. *XV Einstein Telescope Symposium* **27 May 2025**
   Bologna, IT
7. *Institute for Nuclear Theory Workshop 24-89w* **5 Sep 2024**
   University of Washington, US
6. *XIV Einstein Telescope Symposium* **6–7 May 2024**
   Maastricht, NL
5. *SPINS-UK 2023 meeting* **23 Nov 2023**
   Magdalen College, University of Oxford, GB
4. *Institute for Nuclear Theory Program 22-2a* **18 Jul 2022**
   University of Washington, US
3. *PHAROS Conference 2022* **18 May 2022**
   La Sapienza University, Rome, IT
2. *30th Texas Symposium on Relativistic Astrophysics* **17 Dec 2019**
   University of Portsmouth, GB
1. Joint *22nd International Conference on General Relativity and Gravitation* **9 Jul 2019**
   and *13th Edoardo Amaldi Conference on Gravitational Waves*, Valencia, ES

Teaching Experience

Lecturer, Utrecht University, NL **Apr–Jul 2026**

NS-378B, Cosmology
Third-year BSc course with ~ 70 students; mean student evaluation score of $7.6/10$

Instructor, University of Southampton, GB **Feb–May 2024**

MATH1007/1009, Mathematical Methods for Physical Scientists
First-year BSc course with ~ 30 students

Guest Lecturer, University of Southampton, GB

MATH3072, Advanced Fluid Dynamics **Oct 2022, Oct 2023**
Third-year BSc course with ~ 20 students

MATH3006, Relativity, Black Holes and Cosmology **Apr 2022**
Third-year BSc course with ~ 30 students

Teaching Assistant, University of Southampton, GB **Oct 2017–May 2021**

MATH1054/1055, Mathematics for Engineering and the Environment

MATH1057, Dynamics and Relativity

MATH1058, Operational Research I and Mathematical Computing
MATH2045, Vector Calculus and Complex Variable Theory
MATH3018, Numerical Methods
MATH3087, Maths and Your Future

Teaching Assistant, King Edward's School, Birmingham, GB **Jan–Apr 2016**
Physics (11–16 yr)

Mentoring and Supervision

PhD student mentoring

Vincent Ernst, Université Catholique de Louvain, BE **Sep 2025–Present**
Thibeau Wouters, Utrecht University, NL **Oct 2024–Present**
Rahime Matur, University of Southampton, GB **Jan 2023–Sep 2024**
Rhys Counsell, University of Southampton, GB **Sep 2021–Sep 2024**
Shanshan Yin, University of Southampton, GB **Sep 2021–Sep 2024**
Thomas Celora, University of Southampton, GB **Sep 2021–Sep 2023**

BSc student mentoring

Anouk Verburg, Utrecht University, NL **Aug 2026–Present**
Tobie Walraven, Utrecht University, NL **Aug 2025–Jan 2026**

Scientific Collaborations

LIGO-Virgo-KAGRA Collaboration, Member **Oct 2024–Present**
LIGO Collaboration, Member **Oct 2026–Present**
Virgo Collaboration, Member **Oct 2024–Sep 2026**

Cosmic Explorer Consortium, Member **May 2024–Present**

Einstein Telescope Collaboration, Member **Sep 2023–Present**

Leadership and Academic Citizenship

American Physical Society, Member **Oct 2026–Present**
Division of Astrophysics
Division of Gravitational Physics

International Astronomical Union, Junior member **May 2023–Present**

International Society on General Relativity and Gravitation, **May 2021–Present**
Lifetime member

Institute of Physics, Member **Apr 2021–Dec 2025**
Gravitational Physics Group, Committee member **Oct 2021–Sep 2025**

Conference organiser

SPINS-UK 2024 meeting, University of Southampton, GB **10–12 Sep 2024**
Local organising committee, ~ 40 participants

Continuous gravitational waves and neutron stars workshop, **17–20 Jun 2024**
Albert Einstein Institute, Hannover, DE
Scientific organising committee, ~ 50 participants

Gravitational Physics Annual Meeting, Institute of Physics, GB **18 Jan 2024**
Scientific organising committee, ~ 50 participants

23rd BritGrav Conference, University of Southampton, GB **13–14 Apr 2023**
Scientific and local organising committee, ~ 100 participants

Seminar organiser

Gravity Seminar, University of Southampton, GB **Oct 2021–Sep 2024**

Weekly Gravity Reading Group, University of Southampton, GB **Jan–Jul 2021**

Panellist

Fellowships Panel Discussion, GW:UK (online) **26 Jun 2026**

Funding Opportunities Workshop, **27 May 2026**
Gravitational-Wave Early Career Scientists (online)

Professional Service

Journal referee

Astronomy and Astrophysics, Classical and Quantum Gravity, Journal of Cosmology and Astroparticle Physics, Journal of Physics G, Monthly Notices of the Royal Astronomical Society, Nature Astronomy, Physical Review D, Physical Review Letters, The Astrophysical Journal

Project referee

UNamur Postdoctoral Programme, University of Namur, BE **2025**

Open Fellowship, Engineering and Physical Sciences Research Council, GB **2024**

Public Engagement

Outreach activities

Southampton Science and Engineering Festival **7 May 2022, 18 Mar 2023**
Organised neutron-star exhibit for general public and coordinated team of 10 volunteers

Mathematical Challenge **Mar–Apr 2020**
Marked over 200 pupil entries

Maths and Physics Workshop **8 Nov 2017**
Demonstrated for ~ 100 secondary-school pupils

Press (selected)

Lightest neutron star ever found could contain compressed quarks, **24 Oct 2022**
New Scientist

Neutron star 'mountains' may be blocking our view of mysterious gravitational waves, Live Science **21 Jul 2021**

Mountains on neutron stars are not even a millimetre tall due to extreme gravity, The Register **21 Jul 2021**

Scientists find tiny mountains on neutron stars that are a fraction of a millimetre tall, The Independent — **19 Jul 2021**

Neutron Stars Have Mountains That Are Less Than a Millimeter Tall, Gizmodo — **18 Jul 2021**

Neutron stars are remarkably smooth thanks to their intense gravity, New Scientist — **24 May 2021**

Why don't they just break up? Astrobites — **16 Nov 2018**

Computer Skills

Advanced in Julia, Python. Intermediate in Bash, C++, Mathematica, MATLAB. Intermediate in high-performance computing (HTCondor, Slurm). Markup languages: LaTeX, Markdown.

Software—Most contributions can be found at `https://github.com/fgittins`. Member of the *Bilby* development team (`https://github.com/bilby-dev/bilby`). Contributor to *SciML* (`https://sciml.ai`), in particular `NonlinearSolve.jl` (`https://github.com/SciML/NonlinearSolve.jl`). Author of `RealisticSeismology` Julia code (`https://github.com/fgittins/RealisticSeismology`).

References

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